

Majid Daliri

370 Jay St, Brooklyn, NY
11th Floor, Desk A1119

Cell: +1-646-750-4667
daliri.majid@nyu.edu
[majid-daliri.github.io](https://github.com/majid-daliri)

Education	New York University, New York, USA Ph.D. in Computer Science (in progress) Student in the Theoretical Computer Science group Advised by Prof. Christopher Musco	2022 - present
	New York University, New York, USA M.S. in Computer Science	2022 - 2024
	University of Tehran, Tehran, Iran B.S. in Computer Engineering (GPA: 3.97/4.0)	2017 - 2022
Publications	(1) TurboQuant: Online Vector Quantization with Near-optimal Distortion Rate (Arxiv) Amir Zandieh, Majid Daliri, Majid Hadian, Vahab Mirrokni (2) Brains vs. Bytes: Evaluating LLM Proficiency in Olympiad Mathematics (Arxiv) H. Mahdavi, A. Hashemi, Majid Daliri, P Mohammadipour, A. Farhadi, S. Malek, Y. Yazdanifard, A. Khasahmadi, V. Honavar (3) Coupling without Communication and Drafter-Invariant Speculative Decoding (ISIT) 2025 Majid Daliri, Christopher Musco, Ananda Theertha Suresh (4) Unlocking the Theory Behind Scaling 1-Bit Neural Networks (CPAL) 2025 Majid Daliri, Zhao Song, Chiwun Yang (5) QJL: 1-Bit Quantized JL transform for KV Cache Quantization (AAAI) 2025 Amir Zandieh, Majid Daliri, Insu Han (6) Matrix Product Sketching via Coordinated Sampling (ICLR) 2025 Majid Daliri, Juliana Freire, Danrong Li, Christopher Musco (7) Sampling Methods for Inner Product Sketching (VLDB) 2024 Majid Daliri, Juliana Freire, Christopher Musco, Aécio Santos, Haoxiang Zhang (8) Simple Analysis of Priority Sampling (SOSA) 2024 Majid Daliri, Juliana Freire, Christopher Musco, Aécio Santos, Haoxiang Zhang (9) KDEformer: Accelerating Transformers via Kernel Density Estimation (ICML) 2023 Amir Zandieh, Insu Han*, Majid Daliri*, Amin Karbasi (* equal contribution) (10) Weighted Minwise Hashing Beats Linear Sketching for Inner Product Estimation (PODS) 2023 Aline Bessa, Majid Daliri, Juliana Freire, Cameron Musco, Christopher Musco, Aécio Santos, Haoxiang Zhang (11) Efficient Approximations for Cache-conscious Data Placement (PLDI) 2022 Ali Ahmadi, Majid Daliri, Amir Kafshdar Goharshady, Andreas Pavlogiannis (12) A 10-Approximation of the $\frac{\pi}{2}$-MST (STACS) 2022 Ahmad Biniaz, Majid Daliri, AmirHossein Moradpour	
Research Internship	Machine Learning Intern, Max Planck Institute , Advised by Dr. Amir Zandieh - Implemented Fast Attention in Transformers, optimizing accuracy and efficiency in sequence modeling tasks. - Designed an efficient GPU-compatible LSH method, boosting performance in attention approximation. - Technical Stack: BigGAN, PyTorch, Transformer architectures, and advanced sequence modeling tools. Research Intern, HKUST , Advised by Prof. Amir Goharshady - Made a pioneering theoretical contribution to Cache-conscious Data Placement (CDP), addressing a longstanding challenge in optimizing cache hits. - Implemented and tested various cache management policies and algorithms, outperforming previous heuristics. - Our method emerged as the most effective solution, setting a new standard for cache optimization.	2021-2022 2021-2022

	Research Intern, University of Salzburg , Advised by Prof. Ana Sokolova - Developed algorithms for Distribution Bisimilarity, focusing on finite bisimulation up to convex hull witness. - Extended research to probabilistic system verification and collaborated on using Quantum Annealers for program verification.	Summer 2022
Awards and Honors	Apple Scholars in AI/ML Full graduate fellowship for the 2025–2027 academic years.	2025-2027
	Travel Grants Awarded travel grants to present papers at: - AAAI Conference on Artificial Intelligence (2025) - ACM-SIAM Symposium on Discrete Algorithms (2024) - ICERM Fusing Theory and Practice of Graph Algorithms (2025)	2024, 2025
	Research Grant, University of Salzburg Awarded a €5,000 grant for a research internship focusing on algorithms for distribution bisimilarity, probabilistic systems verification, and quantum annealing projects.	Summer 2022
	Hong Kong PhD Fellowship Scheme (declined to attend NYU) Awarded \$185,000 Ph.D. Fellowship as one of the top 300 students selected across all fields for academic excellence and research potential.	2022
	ACM ICPC - Regional (University of Tehran) Ranked 6 th among more than 100 team all around the Iran.	2019
	Iranian National Olympiad in Informatics Finalist (IOI, Iran) Awarded to 50 students after a year long competition involving 10,000 students.	2016
Service	- Reviewer for International Conference on Learning Representations (ICLR 2025) - Reviewer for Conference on Neural Information Processing Systems (NeurIPS 2024) - Reviewer for ACL Rolling Review (ACL, EMNLP, NAACL 2024-2025) - Reviewer for International Conference on Machine Learning (ICML 2024-2025) - Reviewer for Royal Society Open - External Reviewer for Canadian Conference on Computational Geometry (CCCG 2023)	
Conference Presentations	QJL: 1-Bit Quantized JL transform for KV Cache Quantization { Poster, AAAI 2025 Presentation, ICERM 2025 Coupling without Communication Presentation, (NYC TCS Day) 2024 Simple Analysis of Priority Sampling Presentation, (SOSA) 2024 Accelerating Transformers via Kernel Density Estimation Poster, (ICML) 2023 Weighted MinHash for Inner Product Estimation Poster, (PODS) 2023 Efficient Approximations for Cache-conscious Data Placement Presentation, (PLDI) 2022	
Teaching	Section Leader for NYU CSCI-UA 310 Basic Algorithms Spring 2023 Teaching Assistant NYU CS-GY 6763 Algorithmic Machine Learning Fall 2022 Teaching Assistant University of Tehran Design and Analysis of Algorithms Fall 2020-2021	
Work Experience	Site Reliability Engineer at Cafebazaar - Designed, implemented, and maintained both Redis-as-a-Service/PostgreSQL-as-a-Service on a Kubernetes-based cloud. - Achieved consistent performance benchmarks for both services with 100% uptime and a 99.9% response rate. - Technical Stack: Kubernetes, Docker, Sentry, S3, Prometheus	2021 - 2022

Skills

Theoretical Background:

Proficient in Machine Learning Theory, Neural Networks, Linear Algebra, and Probability.

Technical Skills:

Highly skilled in C/C++, CUDA, Go, Python, Bash-Scripting, PHP, JavaScript. Experience with PyTorch, TensorFlow, Django, CSS3, HTML5, and git.